

16 Digital-to-analog converter (DAC1 and DAC2)

16.1 Introduction

The DAC module is a 12-bit, voltage output digital-to-analog converter. The DAC can be configured in 8- or 12-bit mode and may be used in conjunction with the DMA controller. In 12-bit mode, the data could be left- or right-aligned. An input reference voltage, V_{REF+} (shared with ADC), is available. The output can optionally be buffered for higher current drive.

16.2 DAC1/2 main features

The STM32F303xB/C/D/E, STM32F358xC and STM32F398xE integrate two 12-bit DAC channels with output buffer.

The STM32F303x6/8 and STM32F328x8 integrate 3 12-bit DAC channels (1 DAC channel with output buffer and 2 DAC channels without output buffers).

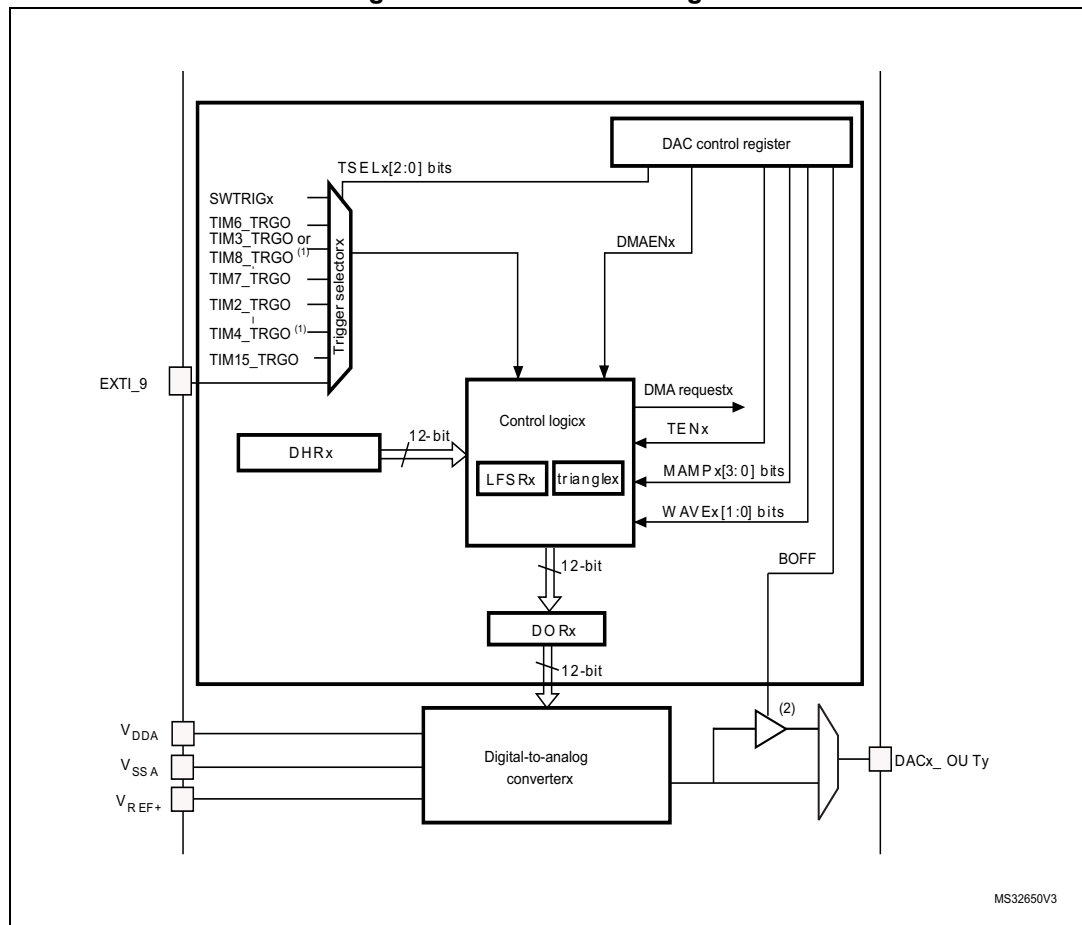
- DAC1 integrates two DAC channels:
 - DAC1 channel 1 which output is DAC1_OUT1
 - DAC1 channel 2 which output is DAC1_OUT2The two channels can be used independently or simultaneously when both channels are grouped together for synchronous update operations (dual mode).
- DAC2 integrates only one channel, DAC2 channel 1 which output is DAC2_OUT1 (STM32F303x6/8 and STM32F328x8 devices only).

The DAC main features are the following:

- Left or right data alignment in 12-bit mode
- Synchronized update capability
- Noise-wave generation (DAC1 only)
- Triangular-wave generation (DAC1 only)
- Independent or simultaneous conversions (dual mode only)
- DMA capability for each channel
- DMA underrun error detection
- External triggers for conversion
- Programmable internal buffer
- Input voltage reference, V_{DDA}

[Figure 113](#) and [Figure 114](#) show the block diagram of a DAC1 and DAC2 channel and [Table 104](#) gives the pin description.

Figure 113. DAC1 block diagram



1. TIM8_TRGO and TIM4_TRGO are only available on STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices.
2. On STM32F303x6/8 and STM32F328, there is no output buffer on the DAC1 channel 2. There is instead a switch allowing to connect the DAC1_OUT2 to the corresponding I/O (PA5) (refer to DAC2 block diagram).

Figure 114. DAC2 block diagram (only on STM32F303x6/8 and STM32F328)

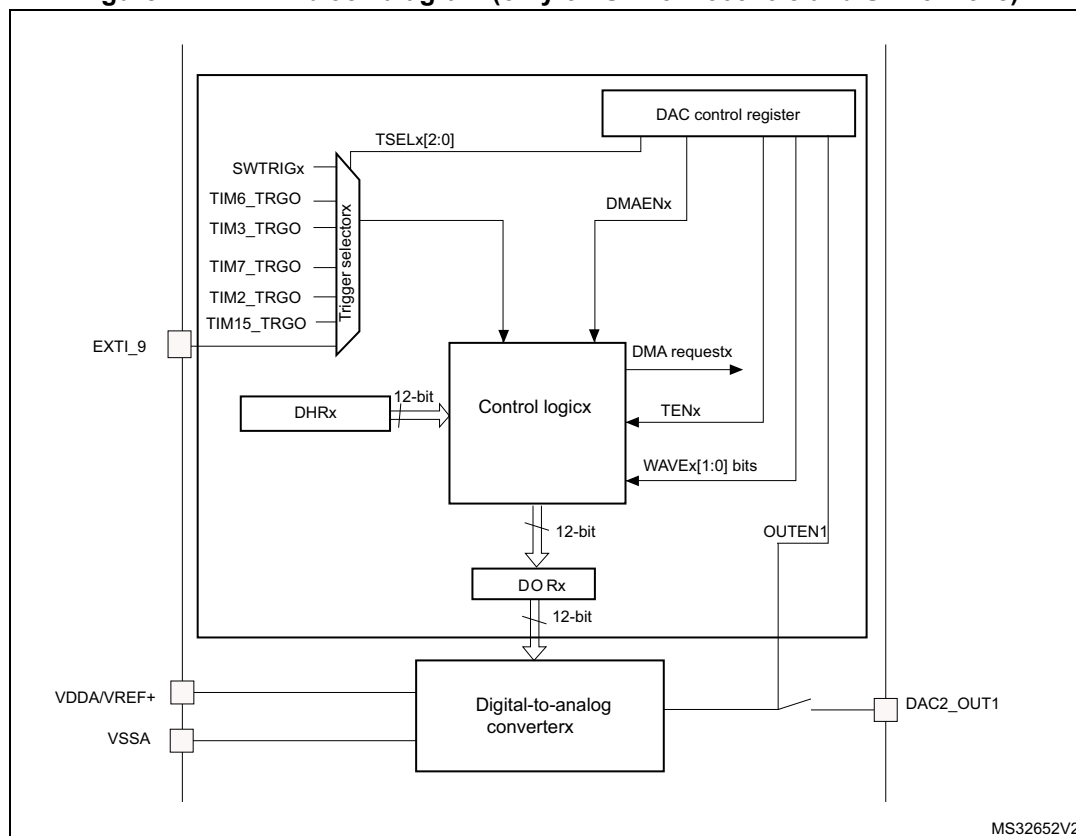


Table 104. DACx pins

Name	Signal type	Remarks
$V_{REF+}^{(1)}$	Input, analog reference positive	The higher/positive reference voltage for the DAC
V_{DDA}	Input, analog supply	Analog power supply
V_{SSA}	Input, analog supply ground	Ground for analog power supply
DAC1_OUT1/2 DAC2_OUT1	Analog output signal	DACx channel y analog output

1. On STM32F303x6/8 and STM32F328, the V_{DDA} and V_{REF+} are internally connected.

Note: Once the DACx channel y is enabled, the corresponding GPIO pin (PA4, PA5 or PA6) is automatically connected to the analog converter output (DACx_OUTy). In order to avoid parasitic consumption, the PA4, PA5 or PA6 pin should first be configured to analog (AIN).

16.3 DAC output buffer enable/DAC output switch

The DAC1 channel 1 and DAC1 channel 2 come with an output buffer that can be used to reduce the output impedance on DAC1_OUT1/2 output, and to drive external loads directly without having to add an external operational amplifier. This feature is available on STM32F303xB/C/D/E, STM32F358xC and STM32F398xE.

In the STM32F303x6/8 and STM32F328, the DAC1 channel 1 comes with an output buffer. The DAC1 channel2 does not have an output buffer, it has instead a switch allowing to connect the DAC1_OUT2 to the corresponding I/O (PA5). The switch can be enabled and disabled through the OUTEN2 bit in the DAC_CR register. The DAC2 channel1 does not have an output buffer, it has instead a switch allowing to connect the DAC2_OUT1 to the corresponding I/O (PA6). The switch can be enabled and disabled through the OUTEN1 bit in the DAC_CR register.

The DAC1 channel output buffer can be enabled and disabled through the BOFF1 bit in the DAC_CR register.

16.4 DAC channel enable

Each DAC channel can be powered on by setting the corresponding ENx bit in the DAC_CR register. Each DAC channel is then enabled after a startup time t_{WAKEUP} .

Note: The ENx bit enables the analog DAC Channelx macrocell only. The DAC Channelx digital interface is enabled even if the ENx bit is reset.

16.5 Single mode functional description

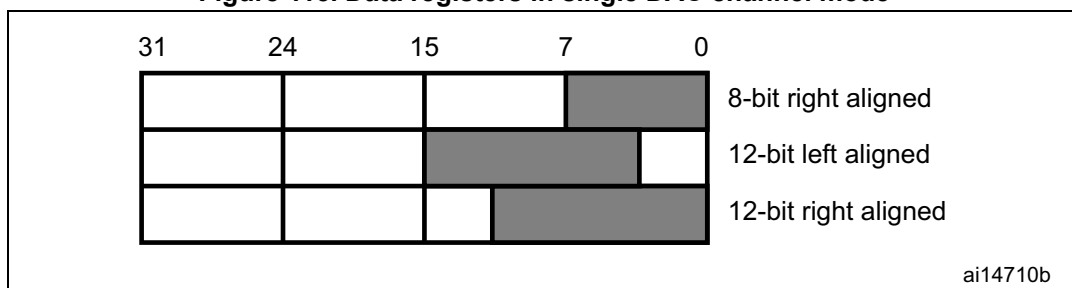
16.5.1 DAC data format

There are three possibilities:

- 8-bit right alignment: the software has to load data into the DAC_DHR8Rx [7:0] bits (stored into the DHRx[11:4] bits)
- 12-bit left alignment: the software has to load data into the DAC_DHR12Lx [15:4] bits (stored into the DHRx[11:0] bits)
- 12-bit right alignment: the software has to load data into the DAC_DHR12Rx [11:0] bits (stored into the DHRx[11:0] bits)

Depending on the loaded DAC_DHRyyyx register, the data written by the user is shifted and stored into the corresponding DHRx (data holding registerx, which are internal non-memory-mapped registers). The DHRx register is then loaded into the DORx register either automatically, by software trigger or by an external event trigger.

Figure 115. Data registers in single DAC channel mode



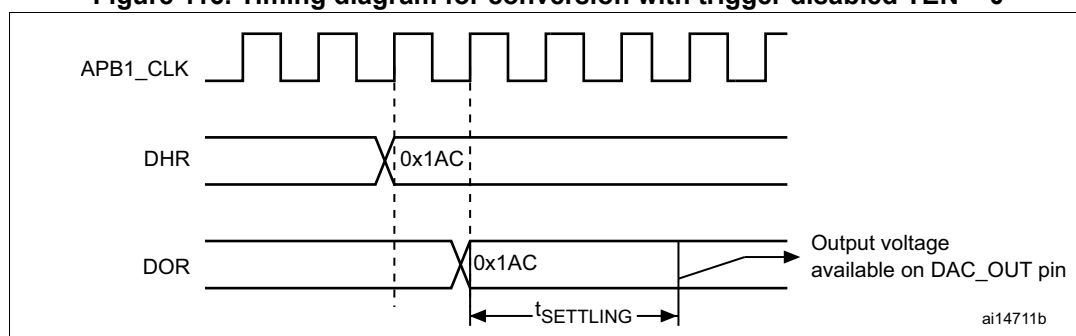
16.5.2 DAC channel conversion

The DAC_DORx cannot be written directly and any data transfer to the DAC channelx must be performed by loading the DAC_DHRx register (write to DAC_DHR8Rx, DAC_DHR12Lx, DAC_DHR12Rx).

Data stored in the DAC_DHRx register are automatically transferred to the DAC_DORx register after one APB1 clock cycle, if no hardware trigger is selected (TENx bit in DAC_CR register is reset). However, when a hardware trigger is selected (TENx bit in DAC_CR register is set) and a trigger occurs, the transfer is performed three PCLK1 clock cycles later.

When DAC_DORx is loaded with the DAC_DHRx contents, the analog output voltage becomes available after a time t_{SETTLING} that depends on the power supply voltage and the analog output load.

Figure 116. Timing diagram for conversion with trigger disabled TEN = 0



Independent trigger with single LFSR generation

To configure the DAC in this conversion mode (see [Section 16.7: Noise generation](#)), the following sequence is required:

1. Set the DAC channel trigger enable bit TENx.
2. Configure the trigger source by setting TSELx[2:0] bits.
3. Configure the DAC channel WAVEx[1:0] bits as "01" and the same LFSR mask value in the MAMPx[3:0] bits
4. Load the DAC channel data into the desired DAC_DHRx register (DHR12RD, DHR12LD or DHR8RD).

When a DAC channelx trigger arrives, the LFSRx counter, with the same mask, is added to the DHRx register and the sum is transferred into DAC_DORx (three APB clock cycles later). Then the LFSRx counter is updated.

Independent trigger with single triangle generation

To configure the DAC in this conversion mode (see [Section 16.8: Triangle-wave generation](#)), the following sequence is required:

1. Set the DAC channelx trigger enable TENx bits.
2. Configure the trigger source by setting TSELx[2:0] bits.
3. Configure the DAC channelx WAVEx[1:0] bits as "1x" and the same maximum amplitude value in the MAMPx[3:0] bits
4. Load the DAC channelx data into the desired DAC_DHRx register. (DHR12RD, DHR12LD or DHR8RD).

When a DAC channelx trigger arrives, the DAC channelx triangle counter, with the same triangle amplitude, is added to the DHRx register and the sum is transferred into DAC_DORx (three APB clock cycles later). The DAC channelx triangle counter is then updated.

16.5.3 DAC output voltage

Digital inputs are converted to output voltages on a linear conversion between 0 and V_{REF+} .

The analog output voltages on each DAC channel pin are determined by the following equation:

$$DAC_{Output} = V_{REF+} \times \frac{DOR}{4096}$$

16.5.4 DAC trigger selection

If the TENx control bit is set, conversion can then be triggered by an external event (timer counter, external interrupt line). The TSELx[2:0] control bits determine which possible events will trigger conversion as shown in [Table 105](#).

Table 105. External triggers (DAC1)

Source	Type	TSEL[2:0]
TIM6_TRGO event	Internal signal from on-chip timers	000
TIM3_TRGO event ⁽¹⁾ or Timer 8 TRGO event ⁽²⁾		001 ⁽²⁾
TIM7_TRGO event		010
TIM15_TRGO event		011
TIM2_TRGO event		100
TIM4_TRGO event		101
EXTI line9	External pin	110
SWTRIG	Software control bit	111

1. To select TIM3_TRGO event as DAC1 trigger source, the DAC_TRIG_RMP bit must be set in SYSCFG_CFGR1 register.
2. When TSEL = 001, the DAC trigger is selected using the DAC_TRIG_RMP bit in the SYSCFG_CFGR1 register. When this bit is cleared, the DAC trigger is the Timer 8 TRGO event. When this bit is set, the DAC trigger is the Timer 3 TRGO event.

Table 106. External triggers (DAC2)

Source	Type	TSEL[2:0]
TIM6_TRGO event	Internal signal from on-chip timers	000
TIM3_TRGO event		001
TIM7_TRGO event		010
TIM15_TRGO event		011
TIM2_TRGO event		100

Table 106. External triggers (DAC2) (continued)

Source	Type	TSEL[2:0]
EXTI line9	External pin	110
SWTRIG	Software control bit	111

Each time a DAC interface detects a rising edge on the selected timer TRGO output, or on the selected external interrupt line 9, the last data stored into the DAC_DHRx register are transferred into the DAC_DORx register. The DAC_DORx register is updated three APB1 cycles after the trigger occurs.

If the software trigger is selected, the conversion starts once the SWTRIG bit is set. SWTRIG is reset by hardware once the DAC_DORx register has been loaded with the DAC_DHRx register contents.

Note: TSELx[2:0] bit cannot be changed when the ENx bit is set. When software trigger is selected, the transfer from the DAC_DHRx register to the DAC_DORx register takes only one APB1 clock cycle.

16.6 Dual-mode functional description

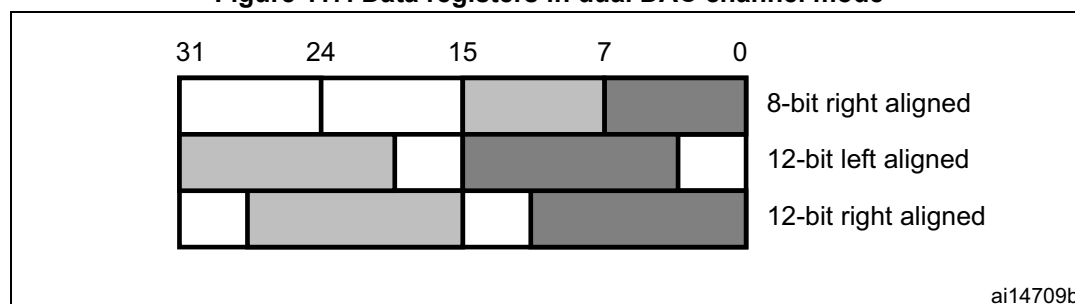
16.6.1 DAC data format

In Dual DAC channel mode, there are three possibilities:

- 8-bit right alignment: data for DAC channel1 to be loaded in the DAC_DHR8RD [7:0] bits (stored in the DHR1[11:4] bits) and data for DAC channel2 to be loaded in the DAC_DHR8RD [15:8] bits (stored in the DHR2[11:4] bits)
- 12-bit left alignment: data for DAC channel1 to be loaded into the DAC_DHR12LD [15:4] bits (stored into the DHR1[11:0] bits) and data for DAC channel2 to be loaded into the DAC_DHR12LD [31:20] bits (stored in the DHR2[11:0] bits)
- 12-bit right alignment: data for DAC channel1 to be loaded into the DAC_DHR12RD [11:0] bits (stored in the DHR1[11:0] bits) and data for DAC channel2 to be loaded into the DAC_DHR12LD [27:16] bits (stored in the DHR2[11:0] bits)

Depending on the loaded DAC_DHRyyyD register, the data written by the user is shifted and stored in DHR1 and DHR2 (data holding registers, which are internal non-memory-mapped registers). The DHR1 and DHR2 registers are then loaded into the DOR1 and DOR2 registers, respectively, either automatically, by software trigger or by an external event trigger.

Figure 117. Data registers in dual DAC channel mode



16.6.2 DAC channel conversion in dual mode

The DAC channel conversion in dual mode is performed in the same way as in single mode (refer to [Section 16.5.2](#)) except that the data have to be loaded by writing to DAC_DHR8Rx, DAC_DHR12Lx, DAC_DHR12Rx, DAC_DHR8RD, DAC_DHR12LD or DAC_DHR12RD.

16.6.3 Description of dual conversion modes

To efficiently use the bus bandwidth in applications that require the two DAC channels at the same time, three dual registers are implemented: DHR8RD, DHR12RD and DHR12LD. A unique register access is then required to drive both DAC channels at the same time.

Eleven conversion modes are possible using the two DAC channels and these dual registers. All the conversion modes can nevertheless be obtained using separate DHRx registers if needed.

All modes are described in the paragraphs below.

Refer to [Section 16.5.2: DAC channel conversion](#) for details on the APB bus (APB or APB1) that clocks the DAC conversions.

Independent trigger without wave generation

To configure the DAC in this conversion mode, the following sequence is required:

1. Set the two DAC channel trigger enable bits TEN1 and TEN2
2. Configure different trigger sources by setting different values in the TSEL1[2:0] and TSEL2[2:0] bits
3. Load the dual DAC channel data into the desired DHR register (DAC_DHR12RD, DAC_DHR12LD or DAC_DHR8RD)

When a DAC channel1 trigger arrives, the DHR1 register is transferred into DAC_DOR1 (three APB clock cycles later).

When a DAC channel2 trigger arrives, the DHR2 register is transferred into DAC_DOR2 (three APB clock cycles later).

Independent trigger with single LFSR generation

To configure the DAC in this conversion mode (refer to [Section 16.7: Noise generation](#)), the following sequence is required:

1. Set the two DAC channel trigger enable bits TEN1 and TEN2
2. Configure different trigger sources by setting different values in the TSEL1[2:0] and TSEL2[2:0] bits
3. Configure the two DAC channel WAVEx[1:0] bits as "01" and the same LFSR mask value in the MAMPx[3:0] bits
4. Load the dual DAC channel data into the desired DHR register (DHR12RD, DHR12LD or DHR8RD)

When a DAC channel1 trigger arrives, the LFSR1 counter, with the same mask, is added to the DHR1 register and the sum is transferred into DAC_DOR1 (three APB clock cycles later). Then the LFSR1 counter is updated.

When a DAC channel2 trigger arrives, the LFSR2 counter, with the same mask, is added to the DHR2 register and the sum is transferred into DAC_DOR2 (three APB clock cycles later). Then the LFSR2 counter is updated.

Independent trigger with different LFSR generation

To configure the DAC in this conversion mode (refer to [Section 16.7: Noise generation](#)), the following sequence is required:

1. Set the two DAC channel trigger enable bits TEN1 and TEN2
2. Configure different trigger sources by setting different values in the TSEL1[2:0] and TSEL2[2:0] bits
3. Configure the two DAC channel WAVEx[1:0] bits as "01" and set different LFSR masks values in the MAMP1[3:0] and MAMP2[3:0] bits
4. Load the dual DAC channel data into the desired DHR register (DAC_DHR12RD, DAC_DHR12LD or DAC_DHR8RD)

When a DAC channel1 trigger arrives, the LFSR1 counter, with the mask configured by MAMP1[3:0], is added to the DHR1 register and the sum is transferred into DAC_DOR1 (three APB clock cycles later). Then the LFSR1 counter is updated.

When a DAC channel2 trigger arrives, the LFSR2 counter, with the mask configured by MAMP2[3:0], is added to the DHR2 register and the sum is transferred into DAC_DOR2 (three APB clock cycles later). Then the LFSR2 counter is updated.

Independent trigger with single triangle generation

To configure the DAC in this conversion mode (refer to [Section 16.8: Triangle-wave generation](#)), the following sequence is required:

1. Set the DAC channelx trigger enable TENx bits.
2. Configure different trigger sources by setting different values in the TSELx[2:0] bits
3. Configure the DAC channelx WAVEx[1:0] bits as "1x" and the same maximum amplitude value in the MAMPx[3:0] bits
4. Load the DAC channelx data into the desired DAC_DHRx register.

Refer to [Section 16.5.2: DAC channel conversion](#) for details on the APB bus (APB or APB1) that clocks the DAC conversions.

When a DAC channelx trigger arrives, the DAC channelx triangle counter, with the same triangle amplitude, is added to the DHRx register and the sum is transferred into DAC_DORx (three APB clock cycles later). The DAC channelx triangle counter is then updated.

Independent trigger with different triangle generation

To configure the DAC in this conversion mode (refer to [Section 16.8: Triangle-wave generation](#)), the following sequence is required:

1. Set the DAC channelx trigger enable TENx bits.
2. Configure different trigger sources by setting different values in the TSELx[2:0] bits
3. Configure the DAC channelx WAVEx[1:0] bits as "1x" and set different maximum amplitude values in the MAMPx[3:0] bits
4. Load the DAC channelx data into the desired DAC_DHRx register.

When a DAC channelx trigger arrives, the DAC channelx triangle counter, with a triangle amplitude configured by MAMPx[3:0], is added to the DHRx register and the sum is transferred into DAC_DORx (three APB clock cycles later). The DAC channelx triangle counter is then updated.

Simultaneous software start

To configure the DAC in this conversion mode, the following sequence is required:

1. Load the dual DAC channel data to the desired DHR register (DAC_DHR12RD, DAC_DHR12LD or DAC_DHR8RD)

In this configuration, one APB clock cycles).

Simultaneous trigger without wave generation

To configure the DAC in this conversion mode, the following sequence is required:

1. Set the two DAC channel trigger enable bits TEN1 and TEN2
2. Configure the same trigger source for both DAC channels by setting the same value in the TSEL1[2:0] and TSEL2[2:0] bits
3. Load the dual DAC channel data to the desired DHR register (DAC_DHR12RD, DAC_DHR12LD or DAC_DHR8RD)

When a trigger arrives, the DHR1 and DHR2 registers are transferred into DAC_DOR1 and DAC_DOR2, respectively (after three APB clock cycles).

Simultaneous trigger with single LFSR generation

To configure the DAC in this conversion mode (refer to [Section 16.7: Noise generation](#)), the following sequence is required:

1. Set the two DAC channel trigger enable bits TEN1 and TEN2
2. Configure the same trigger source for both DAC channels by setting the same value in the TSEL1[2:0] and TSEL2[2:0] bits
3. Configure the two DAC channel WAVEx[1:0] bits as "01" and the same LFSR mask value in the MAMPx[3:0] bits
4. Load the dual DAC channel data to the desired DHR register (DHR12RD, DHR12LD or DHR8RD)

When a trigger arrives, the LFSR1 counter, with the same mask, is added to the DHR1 register and the sum is transferred into DAC_DOR1 (three APB clock cycles later). The LFSR1 counter is then updated. At the same time, the LFSR2 counter, with the same mask, is added to the DHR2 register and the sum is transferred into DAC_DOR2 (three APB clock cycles later). The LFSR2 counter is then updated.

Simultaneous trigger with different LFSR generation

To configure the DAC in this conversion mode (refer to [Section 16.7: Noise generation](#)), the following sequence is required:

1. Set the two DAC channel trigger enable bits TEN1 and TEN2
2. Configure the same trigger source for both DAC channels by setting the same value in the TSEL1[2:0] and TSEL2[2:0] bits
3. Configure the two DAC channel WAVEx[1:0] bits as "01" and set different LFSR mask values using the MAMP1[3:0] and MAMP2[3:0] bits
4. Load the dual DAC channel data into the desired DHR register (DAC_DHR12RD, DAC_DHR12LD or DAC_DHR8RD)

When a trigger arrives, the LFSR1 counter, with the mask configured by MAMP1[3:0], is added to the DHR1 register and the sum is transferred into DAC_DOR1 (three APB clock cycles later). The LFSR1 counter is then updated.

At the same time, the LFSR2 counter, with the mask configured by MAMP2[3:0], is added to the DHR2 register and the sum is transferred into DAC_DOR2 (three APB clock cycles later). The LFSR2 counter is then updated.

Simultaneous trigger with single triangle generation

To configure the DAC in this conversion mode (refer to [Section 16.8: Triangle-wave generation](#)), the following sequence is required:

1. Set the DAC channelx trigger enable TEN1x bits.
2. Configure the same trigger source for both DAC channels by setting the same value in the TSELx[2:0] bits.
3. Configure the DAC channelx WAVEx[1:0] bits as “1x” and the same maximum amplitude value using the MAMPx[3:0] bits
4. Load the DAC channelx data into the desired DAC_DHRx registers.

When a trigger arrives, the DAC channelx triangle counter, with the same triangle amplitude, is added to the DHRx register and the sum is transferred into DAC_DORx (three APB clock cycles later). The DAC channelx triangle counter is then updated.

Simultaneous trigger with different triangle generation

To configure the DAC in this conversion mode ‘refer to [Section 16.8: Triangle-wave generation](#)), the following sequence is required:

1. Set the DAC channelx trigger enable TENx bits.
2. Configure the same trigger source for DAC channelx by setting the same value in the TSELx[2:0] bits
3. Configure the DAC channelx WAVEx[1:0] bits as “1x” and set different maximum amplitude values in the MAMPx[3:0] bits.
4. Load the DAC channelx data into the desired DAC_DHRx registers.

When a trigger arrives, the DAC channelx triangle counter, with a triangle amplitude configured by MAMPx[3:0], is added to the DHRx register and the sum is transferred into DAC_DORx (three APB clock cycles later). Then the DAC channelx triangle counter is updated.

16.6.4 DAC output voltage

Refer to [Section 16.5.3: DAC output voltage](#).

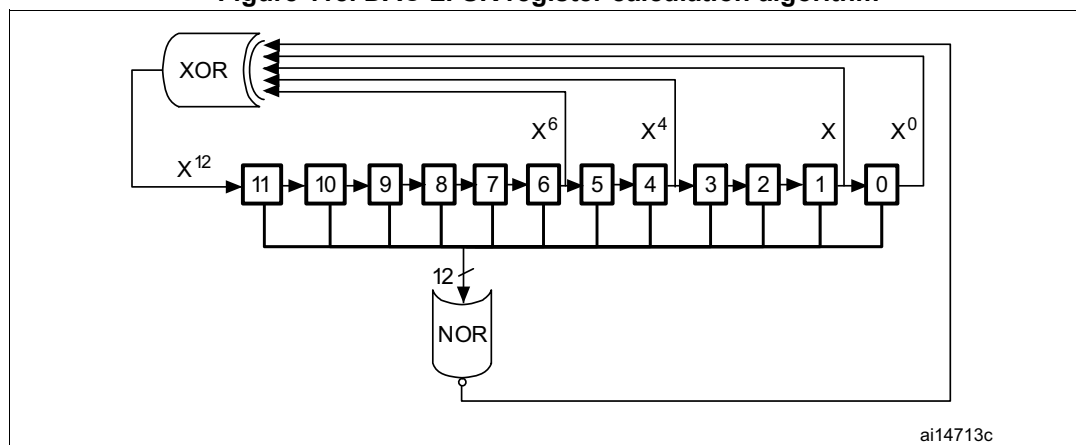
16.6.5 DAC trigger selection

Refer to [Section 16.5.4: DAC trigger selection](#)

16.7 Noise generation

In order to generate a variable-amplitude pseudonoise, an LFSR (linear feedback shift register) is available. DAC noise generation is selected by setting WAVEx[1:0] to "01". The preloaded value in LFSR is 0xAAA. This register is updated three APB clock cycles after each trigger event, following a specific calculation algorithm.

Figure 118. DAC LFSR register calculation algorithm

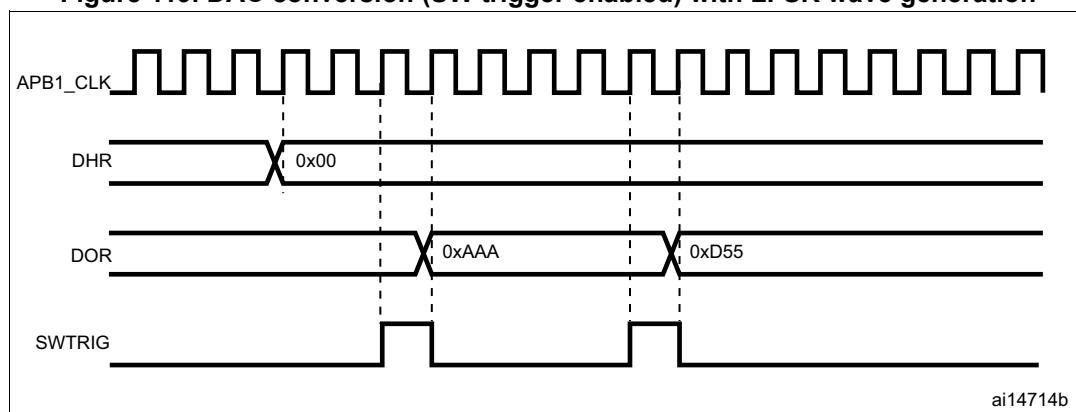


The LFSR value, that may be masked partially or totally by means of the MAMPx[3:0] bits in the DAC_CR register, is added up to the DAC_DHRx contents without overflow and this value is then stored into the DAC_DORx register.

If LFSR is 0x0000, a '1' is injected into it (antilock-up mechanism).

It is possible to reset LFSR wave generation by resetting the WAVEx[1:0] bits.

Figure 119. DAC conversion (SW trigger enabled) with LFSR wave generation



Note: The DAC trigger must be enabled for noise generation by setting the TENx bit in the DAC_CR register.

16.8 Triangle-wave generation

It is possible to add a small-amplitude triangular waveform on a DC or slowly varying signal. DAC triangle-wave generation is selected by setting `WAVEx[1:0]` to “10”. The amplitude is configured through the `MAMPx[3:0]` bits in the `DAC_CR` register. An internal triangle counter is incremented three APB clock cycles after each trigger event. The value of this counter is then added to the `DAC_DHRx` register without overflow and the sum is stored into the `DAC_DORx` register. The triangle counter is incremented as long as it is less than the maximum amplitude defined by the `MAMPx[3:0]` bits. Once the configured amplitude is reached, the counter is decremented down to 0, then incremented again and so on.

It is possible to reset triangle wave generation by resetting the `WAVEx[1:0]` bits.

Figure 120. DAC triangle wave generation

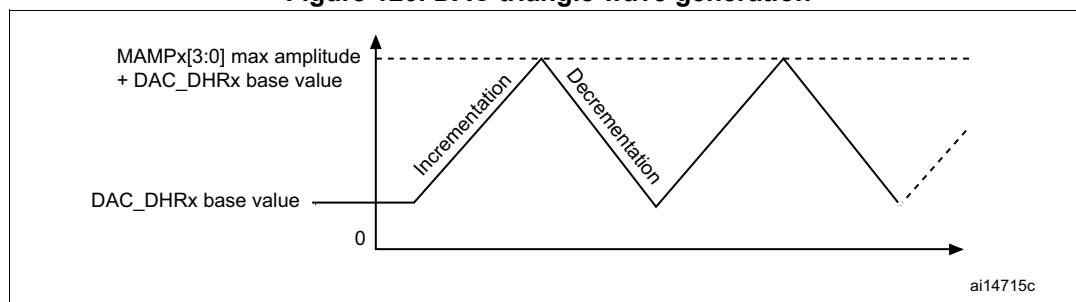
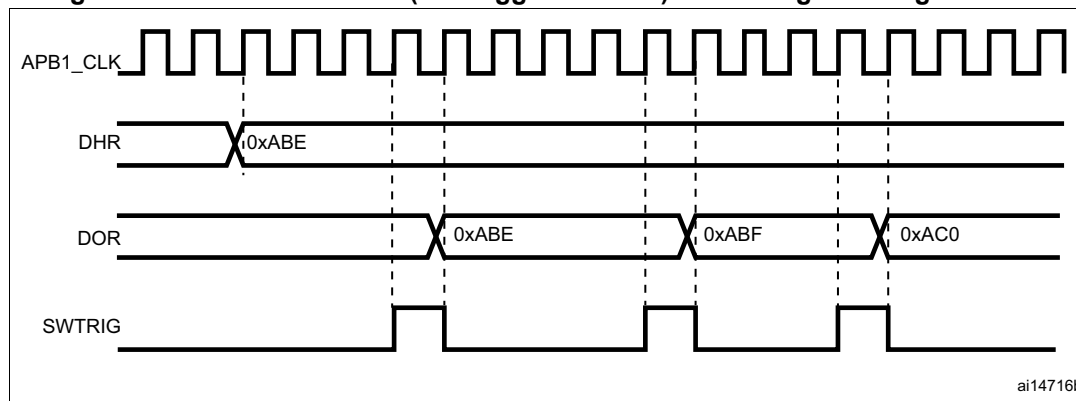


Figure 121. DAC conversion (SW trigger enabled) with triangle wave generation



Note: The DAC trigger must be enabled for triangle generation by setting the `TENx` bit in the `DAC_CR` register.

The `MAMPx[3:0]` bits must be configured before enabling the DAC, otherwise they cannot be changed.

16.9 DMA request

Each DAC channel has a DMA capability. Two DMA channels are used to service DAC channel DMA requests.

A DAC DMA request is generated when an external trigger (but not a software trigger) occurs while the DMAENx bit is set. The value of the DAC_DHRx register is then transferred to the DAC_DORx register.

In dual mode, if both DMAENx bits are set, two DMA requests are generated. If only one DMA request is needed, user should set only the corresponding DMAENx bit. In this way, the application can manage both DAC channels in dual mode by using one DMA request and a unique DMA channel.

DMA underrun

The DAC DMA request is not queued so that if a second external trigger arrives before the acknowledgment for the first external trigger is received (first request), then no new request is issued and the DMA channelx underrun flag DMAUDRx in the DAC_SR register is set, reporting the error condition. DMA data transfers are then disabled and no further DMA request is treated. The DAC channelx continues to convert old data.

The software should clear the DMAUDRx flag by writing “1”, clear the DMAEN bit of the used DMA stream and re-initialize both DMA and DAC channelx to restart the transfer correctly. The software should modify the DAC trigger conversion frequency or lighten the DMA workload to avoid a new DMA. Finally, the DAC conversion can be resumed by enabling both DMA data transfer and conversion trigger.

For each DAC channel, an interrupt is also generated if the corresponding DMAUDRIEx bit in the DAC_CR register is enabled.

16.10 DAC registers

Refer to [Section 2.2 on page 47](#) for a list of abbreviations used in register descriptions.

The peripheral registers have to be accessed by words (32-bit).

16.10.1 DAC control register (DAC_CR)

Address offset: 0x00

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	DMAU DRIE2	DMA EN2	MAMP2[3:0]				WAVE2[1:0]		TSEL2[2:0]			TEN2	BOFF2 /OUTE N2	EN2
		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	DMAU DRIE1	DMA EN1	MAMP1[3:0]				WAVE1[1:0]		TSEL1[2:0]			TEN1	BOFF1 /OUTE N1	EN1
		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **DMAUDRIE2**: DAC channel2 DMA underrun interrupt enable

This bit is set and cleared by software.

0: DAC channel2 DMA underrun interrupt disabled

1: DAC channel2 DMA underrun interrupt enabled

Note: This bit is available in dual mode only. It is reserved in single mode.

Bit 28 **DMAEN2**: DAC channel2 DMA enable

This bit is set and cleared by software.

0: DAC channel2 DMA mode disabled

1: DAC channel2 DMA mode enabled

Note: This bit is available in dual mode only. It is reserved in single mode.

Bits 27:24 **MAMP2[3:0]**: DAC channel2 mask/amplitude selector

These bits are written by software to select mask in wave generation mode or amplitude in triangle generation mode.

0000: Unmask bit0 of LFSR/ triangle amplitude equal to 1

0001: Unmask bits[1:0] of LFSR/ triangle amplitude equal to 3

0010: Unmask bits[2:0] of LFSR/ triangle amplitude equal to 7

0011: Unmask bits[3:0] of LFSR/ triangle amplitude equal to 15

0100: Unmask bits[4:0] of LFSR/ triangle amplitude equal to 31

0101: Unmask bits[5:0] of LFSR/ triangle amplitude equal to 63

0110: Unmask bits[6:0] of LFSR/ triangle amplitude equal to 127

0111: Unmask bits[7:0] of LFSR/ triangle amplitude equal to 255

1000: Unmask bits[8:0] of LFSR/ triangle amplitude equal to 511

1001: Unmask bits[9:0] of LFSR/ triangle amplitude equal to 1023

1010: Unmask bits[10:0] of LFSR/ triangle amplitude equal to 2047

≥1011: Unmask bits[11:0] of LFSR/ triangle amplitude equal to 4095

Note: These bits are available only in dual mode when wave generation is supported.

Otherwise, they are reserved and must be kept at reset value.

- Bits 23:22 **WAVE2[1:0]**: DAC channel2 noise/triangle wave generation enable
 These bits are set/reset by software.
 00: wave generation disabled
 01: Noise wave generation enabled
 1x: Triangle wave generation enabled
Note: Only used if bit TEN2 = 1 (DAC channel2 trigger enabled)
These bits are available only in dual mode when wave generation is supported.
Otherwise, they are reserved and must be kept at reset value.
- Bits 21:19 **TSEL2[2:0]**: DAC channel2 trigger selection
 These bits select the external event used to trigger DAC channel2
 000: Timer 6 TRGO event
 001: Timer 3 or Timer 8 TRGO event depending on the value of DAC_TRIG_RMP bit in SYSCFG_CFGR1 register
 010: Timer 7 TRGO event
 011: Timer 15 TRGO event
 100: Timer 2 TRGO event
 101: Timer 4 TRGO event
 110: EXTI line9
 111: Software trigger
Note: Only used if bit TEN2 = 1 (DAC channel2 trigger enabled).
These bits are available in dual mode only. They are reserved in single mode.
- Bit 18 **TEN2**: DAC channel2 trigger enable
 This bit is set and cleared by software to enable/disable DAC channel2 trigger
 0: DAC channel2 trigger disabled and data written into the DAC_DHRx register are transferred one APB1 clock cycle later to the DAC_DOR2 register
 1: DAC channel2 trigger enabled and data from the DAC_DHRx register are transferred three APB1 clock cycles later to the DAC_DOR2 register
Note: When software trigger is selected, the transfer from the DAC_DHRx register to the DAC_DOR2 register takes only one APB1 clock cycle.
Note: This bit is available in dual mode only. It is reserved in single mode.
- Bit 17 **In STM32F303xB/C/D/E, STM32F358xC and STM32F398xE:**
BOFF2: DAC channel2 output buffer disable
 This bit is set and cleared by software to enable/disable DAC channel2 output buffer.
 0: DAC channel2 output buffer enabled
 1: DAC channel2 output buffer disabled
Note: This bit is available in dual mode only. It is reserved in single mode.
In STM32F303x6/8 and STM32F328x8 DAC1:
OUTEN2: DAC channel2 output switch enable
 This bit is set and cleared by software to enable/disable DAC channel2 output switch.
 0: DAC channel2 output switch disabled
 1: DAC channel2 output switch enabled
- Bit 16 **EN2**: DAC channel2 enable
 This bit is set and cleared by software to enable/disable DAC channel2.
 0: DAC channel2 disabled
 1: DAC channel2 enabled
Note: This bit is available in dual mode only. It is reserved in single mode.
- Bits 15:14 Reserved, must be kept at reset value.

Bit 13 **DMAUDRIE1**: DAC channel1 DMA Underrun Interrupt enable

This bit is set and cleared by software.

- 0: DAC channel1 DMA Underrun Interrupt disabled
- 1: DAC channel1 DMA Underrun Interrupt enabled

Bit 12 **DMAEN1**: DAC channel1 DMA enable

This bit is set and cleared by software.

- 0: DAC channel1 DMA mode disabled
- 1: DAC channel1 DMA mode enabled

Bits 11:8 **MAMP1[3:0]**: DAC channel1 mask/amplitude selector

These bits are written by software to select mask in wave generation mode or amplitude in triangle generation mode.

- 0000: Unmask bit0 of LFSR/ triangle amplitude equal to 1
- 0001: Unmask bits[1:0] of LFSR/ triangle amplitude equal to 3
- 0010: Unmask bits[2:0] of LFSR/ triangle amplitude equal to 7
- 0011: Unmask bits[3:0] of LFSR/ triangle amplitude equal to 15
- 0100: Unmask bits[4:0] of LFSR/ triangle amplitude equal to 31
- 0101: Unmask bits[5:0] of LFSR/ triangle amplitude equal to 63
- 0110: Unmask bits[6:0] of LFSR/ triangle amplitude equal to 127
- 0111: Unmask bits[7:0] of LFSR/ triangle amplitude equal to 255
- 1000: Unmask bits[8:0] of LFSR/ triangle amplitude equal to 511
- 1001: Unmask bits[9:0] of LFSR/ triangle amplitude equal to 1023
- 1010: Unmask bits[10:0] of LFSR/ triangle amplitude equal to 2047
- ≥ 1011: Unmask bits[11:0] of LFSR/ triangle amplitude equal to 4095

Note: These bits are available only when wave generation feature is supported. Otherwise, they are reserved and must be kept at reset value.

Bits 7:6 **WAVE1[1:0]**: DAC channel1 noise/triangle wave generation enable

These bits are set and cleared by software.

- 00: Wave generation disabled
- 01: Noise wave generation enabled
- 1x: Triangle wave generation enabled

Note: Only used if bit TEN1 = 1 (DAC channel1 trigger enabled).

Note: These bits are available only when wave generation feature is supported. Otherwise they are reserved and must be kept at reset value.

Bits 5:3 **TSEL1[2:0]**: DAC channel1 trigger selection

These bits select the external event used to trigger DAC channel1.

- 000: Timer 6 TRGO event
- 001: Timer 3 or Timer 8 TRGO event depending on the value of DAC_TRIG_RMP bit in SYSCFG_CFGR1 register
- 010: Timer 7 TRGO event
- 011: Timer 15 TRGO event
- 100: Timer 2 TRGO event
- 101: Timer 4 TRGO event
- 110: EXTI line9
- 111: Software trigger

Note: Only used if bit TEN1 = 1 (DAC channel1 trigger enabled).

Bit 2 **TEN1**: DAC channel1 trigger enable

This bit is set and cleared by software to enable/disable DAC channel1 trigger.

0: DAC channel1 trigger disabled and data written into the DAC_DHRx register are transferred one APB1 clock cycle later to the DAC_DOR1 register

1: DAC channel1 trigger enabled and data from the DAC_DHRx register are transferred three APB1 clock cycles later to the DAC_DOR1 register

Note: When software trigger is selected, the transfer from the DAC_DHRx register to the DAC_DOR1 register takes only one APB1 clock cycle.

Bit 1 **In DAC1**:**BOFF1**: DAC channel1 output buffer disable

This bit is set and cleared by software to enable/disable DAC channel1 output buffer.

0: DAC channel1 output buffer enabled

1: DAC channel1 output buffer disabled

In DAC2: (STM32F303x6/8 and STM32F328x8 only)**OUTEN1**: DAC channel1 output switch enable

This bit is set and cleared by software to enable/disable DAC channel1 output switch.

0: DAC channel1 output switch disabled

1: DAC channel1 output switch enabled

Bit 0 **EN1**: DAC channel1 enable

This bit is set and cleared by software to enable/disable DAC channel1.

0: DAC channel1 disabled

1: DAC channel1 enabled

16.10.2 DAC software trigger register (DAC_SWTRIGR)

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWTRIG2	SWTRIG1
														w	w

Bits 31:2 Reserved, must be kept at reset value.

Bit 1 **SWTRIG2**: DAC channel2 software trigger

This bit is set and cleared by software to enable/disable the software trigger.

0: Software trigger disabled

1: Software trigger enabled

Note: This bit is cleared by hardware (one APB1 clock cycle later) once the DAC_DHR2 register value has been loaded into the DAC_DOR2 register.

This bit is available in dual mode only. It is reserved in single mode.

Bit 0 **SWTRIG1**: DAC channel1 software trigger

This bit is set and cleared by software to enable/disable the software trigger.

0: Software trigger disabled

1: Software trigger enabled

Note: This bit is cleared by hardware (one APB1 clock cycle later) once the DAC_DHR1 register value has been loaded into the DAC_DOR1 register.

16.10.3 DAC channel1 12-bit right-aligned data holding register (DAC_DHR12R1)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	DACC1DHR[11:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:0 **DACC1DHR[11:0]**: DAC channel1 12-bit right-aligned data

These bits are written by software which specifies 12-bit data for DAC channel1.

16.10.4 DAC channel1 12-bit left-aligned data holding register (DAC_DHR12L1)

Address offset: 0x0C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DAC1DHR[11:0]												v	Res.	Res.	Res.
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw				

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:4 **DACC1DHR[11:0]**: DAC channel1 12-bit left-aligned data

These bits are written by software which specifies 12-bit data for DAC channel1.

Bits 3:0 Reserved, must be kept at reset value.

16.10.5 DAC channel1 8-bit right-aligned data holding register (DAC_DHR8R1)

Address offset: 0x10

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DAC1DHR[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **DACC1DHR[7:0]**: DAC channel1 8-bit right-aligned data

These bits are written by software which specifies 8-bit data for DAC channel1.

16.10.6 DAC channel2 12-bit right-aligned data holding register (DAC_DHR12R2)

Address offset: 0x14

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	DAC2DHR[11:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:0 **DACC2DHR[11:0]**: DAC channel2 12-bit right-aligned data

These bits are written by software which specifies 12-bit data for DAC channel2.

16.10.7 DAC channel2 12-bit left-aligned data holding register (DAC_DHR12L2)

Address offset: 0x18

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DACC2DHR[11:0]												Res.	Res.	Res.	Res.
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw				

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:4 **DACC2DHR[11:0]**: DAC channel2 12-bit left-aligned data

These bits are written by software which specify 12-bit data for DAC channel2.

Bits 3:0 Reserved, must be kept at reset value.

16.10.8 DAC channel2 8-bit right-aligned data holding register (DAC_DHR8R2)

Address offset: 0x1C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DACC2DHR[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **DACC2DHR[7:0]**: DAC channel2 8-bit right-aligned data

These bits are written by software which specifies 8-bit data for DAC channel2.

16.10.9 Dual DAC 12-bit right-aligned data holding register (DAC_DHR12RD)

Address offset: 0x20

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	DACC2DHR[11:0]											
				r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	DACC1DHR[11:0]											
				r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:28 Reserved, must be kept at reset value.

Bits 27:16 **DACC2DHR[11:0]**: DAC channel2 12-bit right-aligned data

These bits are written by software which specifies 12-bit data for DAC channel2.

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:0 **DACC1DHR[11:0]**: DAC channel1 12-bit right-aligned data

These bits are written by software which specifies 12-bit data for DAC channel1.

16.10.10 Dual DAC 12-bit left-aligned data holding register (DAC_DHR12LD)

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DACC2DHR[11:0]												Res.	Res.	Res.	Res.
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DACC1DHR[11:0]												Res.	Res.	Res.	Res.
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w				

Bits 31:20 **DACC2DHR[11:0]**: DAC channel2 12-bit left-aligned data

These bits are written by software which specifies 12-bit data for DAC channel2.

Bits 19:16 Reserved, must be kept at reset value.

Bits 15:4 **DACC1DHR[11:0]**: DAC channel1 12-bit left-aligned data

These bits are written by software which specifies 12-bit data for DAC channel1.

Bits 3:0 Reserved, must be kept at reset value.

16.10.11 Dual DAC 8-bit right-aligned data holding register (DAC_DHR8RD)

Address offset: 0x28

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DACC2DHR[7:0]								DACC1DHR[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:8 **DACC2DHR[7:0]**: DAC channel2 8-bit right-aligned data

These bits are written by software which specifies 8-bit data for DAC channel2.

Bits 7:0 **DACC1DHR[7:0]**: DAC channel1 8-bit right-aligned data

These bits are written by software which specifies 8-bit data for DAC channel1.

16.10.12 DAC channel1 data output register (DAC_DOR1)

Address offset: 0x2C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	DACC1DOR[11:0]											
				r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:0 **DACC1DOR[11:0]**: DAC channel1 data output

These bits are read-only, they contain data output for DAC channel1.

16.10.13 DAC channel2 data output register (DAC_DOR2)

Address offset: 0x30

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	DACC2DOR[11:0]											
				r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:0 **DACC2DOR[11:0]**: DAC channel2 data output

These bits are read-only, they contain data output for DAC channel2.

16.10.14 DAC status register (DAC_SR)

Address offset: 0x34

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	DMAUDR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
		rc_w1													
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	DMAUDR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
		rc_w1													

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **DMAUDR2**: DAC channel2 DMA underrun flag

This bit is set by hardware and cleared by software (by writing it to 1).

0: No DMA underrun error condition occurred for DAC channel2

1: DMA underrun error condition occurred for DAC channel2 (the currently selected trigger is driving DAC channel2 conversion at a frequency higher than the DMA service capability rate)

Note: This bit is available in dual mode only. It is reserved in single mode.

Bits 28:14 Reserved, must be kept at reset value.

Bit 13 **DMAUDR1**: DAC channel1 DMA underrun flag

This bit is set by hardware and cleared by software (by writing it to 1).

0: No DMA underrun error condition occurred for DAC channel1

1: DMA underrun error condition occurred for DAC channel1 (the currently selected trigger is driving DAC channel1 conversion at a frequency higher than the DMA service capability rate)

Bits 12:0 Reserved, must be kept at reset value.

16.10.15 DAC register map

Table 107 summarizes the DAC registers.

Table 107. DAC register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
0x00	DAC_CR	Res	Res	DMAUDRIE2	DMAEN2	MAMP2[3:0]				WAVE2[1:0]		TSEL2[2:0]			TEN2	BOFF2	EN2	Res	Res	DMAUDRIE1	DMAEN1	MAMP1[3:0]				WAVE1[1:0]		TSEL1[2:0]			TEN1	BOFF1	EN1				
	Reset value			0	0	0	0	0	0	0	0	0	0	0	0	0	0			0	0	0	0	0	0	0	0	0	0	0	0	0	0				
0x04	DAC_SWTRIGR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	SWTRIG2	SWTRIG1				
	Reset value																														0	0					
0x08	DAC_DHR12R1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DACC1DHR[11:0]																
	Reset value																				0	0	0	0	0	0	0	0	0	0	0	0	0				
0x0C	DAC_DHR12L1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DACC1DHR[11:0]											Res	Res	Res	Res					
	Reset value																0	0	0	0	0	0	0	0	0	0	0	0	0								
0x10	DAC_DHR8R1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DACC1DHR[7:0]											
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
0x14	DAC_DHR12R2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DACC2DHR[11:0]																
	Reset value																				0	0	0	0	0	0	0	0	0	0	0	0	0				
0x18	DAC_DHR12L2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DACC2DHR[11:0]											Res	Res	Res			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0								
0x1C	DAC_DHR8R2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DACC2DHR[7:0]											
	Reset value																								0	0	0	0	0	0	0	0	0				
0x20	DAC_DHR12RD	Res	Res	Res	Res	DACC2DHR[11:0]											Res	Res	Res	Res	DACC1DHR[11:0]																
	Reset value					0	0	0	0	0	0	0	0	0	0	0	0					0	0	0	0	0	0	0	0	0	0	0	0				
0x24	DAC_DHR12LD	DACC2DHR[11:0]											Res	Res	Res	Res	DACC1DHR[11:0]											Res	Res	Res	Res						
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0					0	0	0	0	0	0	0	0	0	0	0	0								
0x28	DAC_DHR8RD	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DACC2DHR[7:0]							DACC1DHR[7:0]												
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
0x2C	DAC_DOR1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DACC1DOR[11:0]																
	Reset value																				0	0	0	0	0	0	0	0	0	0	0	0	0				
0x30	DAC_DOR2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DACC2DOR[11:0]																
	Reset value																				0	0	0	0	0	0	0	0	0	0	0	0	0				

Table 107. DAC register map (continued)and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x34	DAC_SR	Res.	Res.	DMAUDR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DMAUDR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value			0																0													

Refer to [Section 3.2 on page 53](#) for the register boundary addresses.